

## EDUCATION

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### North Carolina State University

*Ph.D. Computer Science*

Raleigh, NC

*Aug 2019 - Jul 2024*

- Advisor: *Dr. Tim Menzies*

- Dissertation: [Guidelines for the Application of Neural Technologies in Software Analytics \(or: How to Do More with Less in SE\)](#)

### PES University

*B.E. Computer Science – Advisor: Dr. Snehanshu Saha*

Bangalore, India

*Aug 2015 - May 2019*

## EMPLOYMENT

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### LexisNexis Legal & Professional

*Senior Data Scientist I*

Raleigh, NC

*May 2024 - Present*

- Led the migration from RAG to an agentic legal drafting system as tech lead, and own its continued development.
- Reduced customer-facing product runtime by 24.8% and cut time spent at peak memory from 86.4% to 68% via lazy module loading, LLM call profiling, and parallelization.
- As part of a 3-person team, improved complaint drafting usefulness (as rated by attorney reviewers) from 40% to 86% and motion drafting from 59.2% to 76%.
- Developed a fast (median 12ms) passage filtering approach based on contrastive loss.
- Mentored an intern on a large-scale, full factorial experiment investigating Pareto-optimal configurations for draft quality and latency.
- Built team developer tooling: a Rust MITM proxy with request replay, a web UI for visualizing intermediate agent steps, and custom Python lint rules beyond `ruff`'s rule set.
- **Technology:** Python, Litestar, Google ADK, React, Rust, TypeScript, Tailwind

### North Carolina State University

*PhD Student*

Raleigh, NC

*Aug 2019 - Jul 2024*

- **State-of-the-art hyper-parameter optimization:** Proposed a novel hyper-parameter optimization method that outperforms prior SOTA and is 36.4% faster.
- **Better, faster deep learning for SE:** Improved defect prediction by up to 123% (F-1 score), code smell detection by up to 30% (AUC), issue lifetime prediction by up to 76% (accuracy), automated microservice partitioning by up to 285% (modularity) compared to prior SOTA.
- **Semi-supervised learning:** Achieved state-of-the-art results (up to 100% improvement in AUC) on static code warnings analysis using 10% of the labels.
- **Teaching:** TA for 830 students over 5 semesters across CSC 230 (C and Software Tools), CSC 510 (Software Engineering), and CSC 591/791 (Automated Software Engineering)

### Amazon

*Software Dev Engineer Intern*

New York, NY / Bellevue, WA

*May 2023 - Aug 2023*

- Implemented and shipped profile locks for Prime Video on Echo Show devices.
- **Technology:** React Native, TypeScript

*Software Dev Engineer Intern*

*May 2022 - Jul 2022*

- Developed a full-stack system to publish announcements in scorecards used by delivery service partners.
- **Technology:** React/Redux, TypeScript, Redux Saga, DynamoDB, Java Spring

## RECENT PUBLICATIONS

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See full list on [Google Scholar](#).

1. **Yedida, R.**, & Menzies, T. (2025). Is Hyper-Parameter Optimization Different for Software Analytics? *IEEE Transactions on Software Engineering*, 51(6).
2. Baldassarre, M. T., Ernst, N., Hermann, B., Menzies, T., & **Yedida, R.** (2023). (Re)use of Research Results (is Rampant). *Communications of the ACM*, 66(2), 75-81.
3. **Yedida, R.**, Kang, H. J., Tu, K., Lo, D., & Menzies, T. (2023). How to Find Actionable Static Analysis Warnings: A Case Study with FindBugs. *IEEE Transactions on Software Engineering*, 49(4), 2856-2872.

4. **Yedida, R.**, Krishna, R., Kalia, A., Menzies, T., Xiao, J., & Vukovic, M. (2023). An Expert System for Redesigning Software for Cloud Applications. *Expert Systems with Applications*, 219, 119673.
5. **Yedida, R.**, Menzies, T. (2022). How to Improve Deep Learning for Software Analytics (a case study with code smell detection). In *2022 IEEE/ACM 19th International Conference on Mining Software Repositories (MSR)*.
6. **Yedida, R.**, & Menzies, T. (2021). On the Value of Oversampling for Deep Learning in Software Defect Prediction. *IEEE Transactions on Software Engineering*, 48(8), 3103-3116.
7. Agrawal, A., Yang, X., Agrawal, R., **Yedida, R.**, Shen, X., & Menzies, T. (2021). Simpler Hyperparameter Optimization for Software Analytics: Why, How, When?. *IEEE Transactions on Software Engineering*, 48(8), 2939-2954.
8. Yang, X., Chen, J., **Yedida, R.**, Yu, Z., & Menzies, T. (2021). Learning to recognize actionable static code warnings (is intrinsically easy). *Empirical Software Engineering*, 26(3), 1-24.
9. **Yedida, R.**, Krishna, R., Kalia, A., Menzies, T., Xiao, J., & Vukovic, M. (2021). Lessons learned from hyper-parameter tuning for microservice candidate identification. *Proceedings of the thirty-sixth IEEE/ACM International Conference on Automated Software Engineering (ASE)*.
10. **Yedida, R.**, & Saha, S. (2021). Beginning with Machine Learning: A Comprehensive Primer. *The European Physical Journal Special Topics*, 230(10), 2363-2444.
11. Saha, S., Nagaraj, N., Mathur, A., **Yedida, R.**, & Sneha, H. R. (2020). Evolution of novel activation functions in neural network training for astronomy data: habitability classification of exoplanets. *The European Physical Journal Special Topics*, 229(16), 2629-2738.
12. **Yedida, R.**, Saha, S., & Prashanth, T. (2020). LipschitzLR: Using theoretically computed adaptive learning rates for fast convergence. *Applied Intelligence*, 1-19.
13. Sridhar, S., Saha, S., Shaikh, A., **Yedida, R.**, & Saha, S. (2020, July). Parsimonious Computing: A Minority Training Regime for Effective Prediction in Large Microarray Expression Data Sets. In *2020 International Joint Conference on Neural Networks (IJCNN)* (pp. 1-8).
14. Khaidem, L., **Yedida, R.**, & Theophilus, A. J. (2019, November). Optimizing Inter-nationality of Journals: A Classical Gradient Approach Revisited via Swarm Intelligence. In *International Conference on Modeling, Machine Learning and Astronomy* (pp. 3-14). *Springer, Singapore*.

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## PREPRINTS

1. Hearn, T., Evans, W., **Yedida, R.**, Abdelshiheed, M., Rebei, A., & Khazaeli, S. (2025). Automated Legal Brief Drafting via Iterative Retrieval-Augmented Generation.
2. Rashid, P., Pal, S., Samudrala, V. K., **Yedida, R.**, & Gehringer, E. (2025). Enhancing Peer Assessment Reliability: A Probabilistic Approach Integrating Formative and Summative Feedback.
3. Mehendale, S., Challa, A., **Yedida, R.**, Danda, S., Sarkar, S., & Saha, S. (2025). Radon–Nikodym Derivative: Re-imagining Anomaly Detection from a Measure Theoretic Perspective.
4. **Yedida, R.**, & Saha, S. (2024). Strong convexity-guided hyper-parameter optimization for flatter losses.

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## FUNDING

**\$5,000**, Google Cloud Academic Research Grant, Feb 2022

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## SERVICE TO PROFESSION

**Contributing Member**, Python Software Foundation (Mar 2021 - Present)

**Guest Editor**, IEEE Software SI: The Impact of AI on Productivity and Code 2025; ASEj SI: Replications and Negative Results (RENE) 2025; EMSE SI: Replications and Negative Results (RENE) 2025

**Workshop Co-Chair**, Intelligent Software Engineering (ASE '25, ICST '26); Replications and Negative Results (ASE '24)

**PC Member**, AAAI 2025-2027; FSE 2026-2027; ICSE 2026-2027; EDM 2026; ASE Industry Track 2026; AI Foundation Models and Software Engineering (FORGE) 2024; ICSME Artifact Evaluation Track, 2021-2023; ASE Artifact Evaluation Track, 2022; International Conference on Modeling, Machine Learning, and Astronomy (MMLA), 2019

**Reviewer**, TOSEM 2026; TSE 2025-2026; NeurIPS 2023, 2025; EMSE 2021, 2025; ASEj 2025; ICML 2024-2025; ICLR 2024-2025; NCAA 2023-2025; TMLR 2024; Neural Processing Letters 2023-2024; Artificial Intelligence Review 2023; J. Big Data, 2023; ASE 2023; IEEE SSCI 2020

## HONORS

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|-----------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| <b>Distinguished Reviewer, ICSE 2026</b>                                                                                                | Mar 2026            |
| <b>Google Developer Expert - ML and GCP</b>                                                                                             | Mar 2025 - Present  |
| A global network of experts recognized by Google for deep technical expertise, community leadership, and mentorship.                    |                     |
| <b>Google Cloud Champion Innovator - Cloud AI/ML</b>                                                                                    | Oct 2022 - Mar 2025 |
| A Googler-nominated global network of 500+ technical experts in Google Cloud; merged into the Google Developer Experts program in 2025. |                     |
| <b>Google Cloud Research Innovator</b>                                                                                                  | Feb 2022            |
| Research Innovators are a global community of researchers driving scientific breakthroughs with Google Cloud.                           |                     |

## INVITED TALKS

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**Feb 2024**, “*Improving deep learning performance using theoretical ML*” at BITS Pilani, KK Birla Goa Campus, India

## RELEVANT PROJECTS

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| <b>pi-permission-system (maintained fork)</b><br><i>TypeScript, Node.js</i>                                                                                                                                                                                                                                                    | May 2026 - Present<br><a href="#">GitHub</a>                                                            |
| Maintain a permission-system fork for the Pi coding agent. Redesigned shell permissions using grammar-based analysis of pipelines, substitutions, and redirections, with protected-path checks and argument-aware approval rules. Extended enforcement to background commands and fixed concurrent subagent approval handling. |                                                                                                         |
| <b>Q&amp;A System for Zotero</b><br><i>Rust, SQLite, LanceDB</i>                                                                                                                                                                                                                                                               | Oct 2024 - Present<br><a href="#">GitHub</a> :: <a href="#">crates.io</a>                               |
| Built an academic question-answering and semantic search system over a user's Zotero library. Wrote a custom PDF parser for academic papers (p50 $\approx$ 2,400 pages/s, p90 $\approx$ 900 pages/s; M4 Pro) and a RAG library with support for multiple providers and tool calling.                                           |                                                                                                         |
| <b>JournalBear</b><br><i>TypeScript, React, Zustand, Electron, SwiftUI</i>                                                                                                                                                                                                                                                     | Jan 2017 - Jul 2026<br><a href="#">GitHub</a> :: <a href="#">App Store</a> :: <a href="#">Softpedia</a> |
| Cross-platform journaling app with AES-256 encryption and separate native apps for iOS and macOS. Rated 4/5 by Softpedia.                                                                                                                                                                                                      |                                                                                                         |
| <b>The 1 Billion Row Challenge (1BRC)</b><br><i>Zig</i>                                                                                                                                                                                                                                                                        | Feb 2026 - Apr 2026<br><a href="#">GitHub</a>                                                           |
| Wrote a Zig implementation for the 1BRC, running in 1.20s (M4 Pro, 14 core) using mmap, open addressing (Wyhash + Robin Hood probing), and packed SIMD.                                                                                                                                                                        |                                                                                                         |
| <b>RAISE</b><br><i>Python, Keras</i>                                                                                                                                                                                                                                                                                           | Aug 2020 - Jun 2025<br><a href="#">GitHub</a> :: <a href="#">PyPI</a>                                   |
| Sole developer of an ML utilities Python package used by multiple research groups for reproducible experiments. Downloaded 74k+ times on PyPI.                                                                                                                                                                                 |                                                                                                         |
| <b>Programmable Resumes</b><br><i>Python</i>                                                                                                                                                                                                                                                                                   | Aug 2023 - Mar 2025<br><a href="#">GitHub</a>                                                           |
| Developed a specification and implementation for modular, customizable resumes with support for two popular LaTeX templates. Wrote this resume using this tool.                                                                                                                                                                |                                                                                                         |
| <b>pysh</b><br><i>C++, TMLanguage</i>                                                                                                                                                                                                                                                                                          | May 2021 - Dec 2024<br><a href="#">GitHub</a>                                                           |
| Developed a superset of Python that allows running shell code natively, with a VS Code syntax highlighting extension.                                                                                                                                                                                                          |                                                                                                         |
| <b>Threaded Discussions Website</b><br><i>MongoDB, Node.js, React</i>                                                                                                                                                                                                                                                          | Feb 2021 - Jun 2021<br><a href="#">GitHub</a>                                                           |
| Built a companion site for video calls supporting Reddit-style threaded discussions.                                                                                                                                                                                                                                           |                                                                                                         |
| <b>Google/Meta Data Mining</b><br><i>Python, Keras</i>                                                                                                                                                                                                                                                                         | Feb 2021 - May 2021<br><a href="#">GitHub</a>                                                           |
| Mined Google Takeout and Meta data exports to predict which Amazon best-sellers to advertise to a user (DistilGPT-2, 0.6 F-1 score).                                                                                                                                                                                           |                                                                                                         |
| <b>NearConnect</b><br><i>iOS, SwiftUI</i>                                                                                                                                                                                                                                                                                      | Nov 2020 - Mar 2021<br><a href="#">GitHub</a> :: <a href="#">App Store</a>                              |
| iOS app to connect with people nearby using multicast peer-to-peer connections.                                                                                                                                                                                                                                                |                                                                                                         |

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|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|
| <b>Novel Drug Repurposing Hypotheses</b><br><i>Python, PyTorch</i><br>Identified novel drug repurposing hypotheses by text mining transcripts of an NPR medical radio show, and verified results using a knowledge graph.                                                                                           | Oct 2019 - Feb 2020<br><a href="#">GitHub</a>    |
| <b>Personalized Chatbot</b><br><i>Python, Keras</i><br>Fine-tuned a GPT-2 345M model on 730k messages from Telegram logs to create a personalized chatbot.                                                                                                                                                          | May 2019 - May 2019<br><a href="#">GitHub</a>    |
| <b>Intelligent Tutoring System</b><br><i>Python</i><br>Implemented an Intelligent Tutoring System backend using Bayesian Knowledge Tracing and a novel question selection algorithm.                                                                                                                                | Sep 2018 - May 2019<br><a href="#">GitHub</a>    |
| <b>Astronomy Image Restoration</b><br><i>Python, Keras</i><br>Developed a machine learning approach to restore astronomical images affected by PSF anisotropy and smearing in crowded-field photometry. Improved data retention and accuracy in differential imaging analysis of long-baseline optical time series. | Aug 2018 - Nov 2018<br><a href="#">GitHub</a>    |
| <b>Human Activity Data Project</b><br><i>Python, Keras</i><br>Collected personal activity data for 9 months, grouped tasks into 21 categories. Analyzed most productive hours of the day.                                                                                                                           | Oct 2018 - Nov 2018<br><a href="#">GitHub</a>    |
| <b>Results Scraper</b><br><i>MongoDB, Express.js, React, Node.js</i><br>Website for scraping university examination results and displaying charts and printable reports, with caching.                                                                                                                              | Mar 2018 - Aug 2018<br><a href="#">GitHub</a>    |
| <b>Video Sharing Website</b><br><i>MySQL, Express.js, React, Node.js, Sass, Elasticsearch</i><br>Simplified implementation of a video-sharing website with subscriptions and custom searching.                                                                                                                      | Oct 2017 - Dec 2017<br><a href="#">GitHub</a>    |
| <b>Web development projects</b><br><i>MongoDB, Express.js, React, Node.js, Sass, D3.js</i><br>Projects include URL shortener, rogue-like dungeon crawler game, voting application, Simon game, land surface temperature heatmap, and mapping meteorite impacts across the globe.                                    | Jan 2017 - Nov 2017<br><a href="#">CodePen</a>   |
| <b>Xtreme Calculations</b><br><i>VB.NET, Python</i><br>Windows math software to solve scientific and mathematical problems, with over 30k downloads across multiple sites.                                                                                                                                          | Apr 2013 - Oct 2017<br><a href="#">Softpedia</a> |
| <b>Video Indexer</b><br><i>C++, Qt, Sphinx</i><br>Cross-platform desktop application to detect the time(s) a given keyword was spoken in a video.                                                                                                                                                                   | Jun 2017 - Sep 2017<br><a href="#">GitHub</a>    |

## SKILLS

**Languages:** Rust, Python, TypeScript, Java, C++, Gleam

**ML/AI:** Keras, PyTorch, Hyper-parameter optimization, RAG, LLM evaluation, Agentic systems, Google ADK

**Frameworks:** React, Flask, Litestar, FastAPI

**Databases:** LanceDB, MySQL, MongoDB, DynamoDB

**Cloud:** Google Cloud (Compute Engine, Cloud Storage, Vertex AI), AWS (EC2, S3, DynamoDB)